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ORIGINAL CONTRIBUTION

A double exponential chirp waveform for noisy rheology

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Abstract
In the search for faster rheometrical measurements techniques for fast time-evolving systems, optimally windowed chirps (OWCh) have recently been proposed for the determination of the complex modulus. However, such chirps are prone to artefacts at high frequencies due to fact that the input power is distributed over a range of frequencies leading to reduced signal-to-noise ratios in noisy conditions. The Tukey window which modulates the amplitude of the excitation disturbance and which is required to avoid spectral leakage directly reduces the signal-to-noise ratio at the edges of the signal leading to a divergence of the measured moduli at high frequencies. A new double exponential chirp (DEC) signal is proposed to overcome these limitations. Its capabilities are demonstrated with orthogonal superposition rheometry as an example of a demanding high-noise environment. The S-shaped time-frequency history of the new chirp signal redistributes the input power over the frequency spectrum. Numerical simulations using the Maxwell and Giesekus models, along with orthogonal superposition measurements on wormlike micellar fluids, demonstrate the effectiveness of the DEC waveform. Parameter optimization with the Giesekus model identifies the ideal input configurations for achieving a maximum signal-to-noise ratio during rheological measurements.

Keywords Orthogonal superposition · Chirp signals · Time-resolved rheometry · Wormlike micelles · Giesekus model

Introduction

Oscillatory frequency sweeps are one of the most prevalent experimental techniques in rheology and enable the rheologist to provide key insights into the microstructural behaviour of materials across various timescales. For the past few decades, the accepted experimental methodology has essentially remained unchanged. By applying consecutive single-frequency sinusoidal strain (or stress) forcing to drive deformation in the material under investigation, the amplitude and phase of the resulting shear stress (or strain) are measured simultaneously. The storage and loss moduli are subsequently calculated by processing the measured data using either cross-correlation in the time domain or a Fourier transform algorithm to provide spectral resolution in the frequency domain. The main disadvantages of using such discrete frequency sweeps (DFS) are the loss of information on time localization and the exponential increase in experiment duration when acquiring data at lower frequencies. The latter constraint especially limits the applicability of DFS, as it requires time-translation invariance, in essence that the material under investigation does not mutate significantly within the experimental timeframe. To characterize the temporal change of a material over the required experimental duration, Winter et al. (1988) and Mours and Winter (1994) defined the mutation number

$$Mu = \Delta t_{exp}/\tau_{mu} \quad ; \quad \tau_{mu} = \left(\frac{1}{G}\frac{\partial G}{\partial t}\right)\Big|_{\omega}, \tag{1}$$

where Δt is the experimental observation time and τ_{mu} is the characteristic mutation time of the sample. Only for small mutation numbers $Mu \ll 1$ can changes in the material properties during the measurement be neglected so that the results obtained with the DFS are valid. Fast time-evolving systems are, by definition, excluded from using DFS as a valid measuring method.

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One solution towards the measurement of quickly mutating materials is the multiwave method (Holly et al. 1988) in which the strain input is composed of a superposition of multiple sinusoidal signals:

$$\gamma = \sum_{i=1}^{m} \gamma_i \sin \omega_i t \tag{2}$$

with m being the number of harmonics, ω_i the angular frequency of the i-th harmonic and t the time. However, this method has limitations if the experiment needs to stay within the linear viscoelastic regime of the material: As pointed out by Ghiringhelli et al. (2012), due to linear superposition and constructive interference, the maximum strain amplitude increases proportionally with the number of harmonics multiplied by the order of magnitude of the fundamental harmonic. An added drawback is that the total test duration required Δt_{exp} is still typically about five times the period of the lowest frequency.

Ghiringhelli et al. (2012) proposed the use of "chirp" or "swept sine" signals, characterized by a continuously changing time-frequency spectrum, as an alternative solution. The frequency of the aforementioned chirp signal typically increases exponentially with time and the resulting strain input is described by the following:

$$\gamma(t) = \gamma_0 \sin\left[\frac{\omega_1 T}{\ln W}\left(W^{t/T} - 1\right)\right] \quad ; \quad W = \frac{\omega_2}{\omega_1} \tag{3}$$

with ω_1 and ω_2 being the start and end frequency, respectively, and T representing the total duration of the chirp signal. As this signal is considered to be both time- and Fourier optimal, the minimal signal duration can be set to $T_{min} = 2\pi/\omega_1$, which corresponds to the frequency resolution of the discrete Fourier transform. Besides reducing the total experiment duration, chirps provide both time and frequency resolution as the chirp pulses are localized in time. Because of the short duration of a single chirp, typically less than a minute, the resulting frequency sweep can be considered to be a characteristic snapshot of the material at that time instance, assuming the material does not mutate significantly within the timeframe T_{min} of the chirp signal. To further enhance the signal quality obtained from the chirp pulse, Geri et al. (2018) defined an optimally windowed chirp (OWCh) by modulating the strain amplitude of the chirp pulse in (3) using a Tukey window:

$$h(t) = \begin{cases} \cos^2\left[\frac{\pi}{r}\left(\frac{t}{T} - \frac{r}{2}\right)\right] & \text{if } \frac{t}{T} \le \frac{r}{2} \\ 1 & \text{if } \frac{r}{2} < \frac{t}{T} < 1 - \frac{r}{2} \\ \cos^2\left[\frac{\pi}{r}\left(\frac{t}{T} - 1 + \frac{r}{2}\right)\right] & \text{if } \frac{t}{T} \ge \frac{r}{2} \end{cases} \tag{4}$$

The resulting optimized signal is thus given by $\gamma_{owch}(t) = h(t)\gamma(t)$. Residual oscillations due to spectral leakage are suppressed by the addition of the Tukey window, for which the tapering severity is determined by the Tukey parameter r. If the value of r is zero, the Tukey window simplifies to a rectangular window. With increasing values of r, the amplitude of the chirp is modulated and the edges of the window become more tapered.

Chirps have already been used successfully to trace the gelation of different systems (Ghiringhelli et al. 2012; Curtis et al. 2015; Keshavarz et al. 2021) as well as to study the ageing behaviour of bentonite suspensions (Rathinaraj et al. 2023) and the chemical crosslinking of epoxies (Athanasiou et al. 2024). Building on the chirp concept, Rathinaraj et al. (2022) recently combined the OWCh signal with orthogonal superposition (OSP) rheometry whereby a chirp waveform is superimposed in a mutually perpendicular direction to a steady shearing flow. This orthogonality of the steady shear flow and the chirp input minimizes the nonlinear coupling of the two flow fields compared to parallel superposition, in which both inputs are applied in the same direction (Vermant et al. 1998). Aptly named "OrthoChirp", this method was utilized to probe how the evolution of the linear viscoelastic properties and relaxation spectra of alginate gels were modified by a strong shear flow.

However, particularly in environments with considerable levels of instrumentation noise, such as the case of OSP measurements, a previously unknown artefact emerged. In Fig. 1, multiple orthogonal small amplitude oscillatory shear (SAOS) frequency sweeps on different Newtonian Brookfield standards are shown, representing a limiting case of OSP measurements with the steady shear rate $\dot{\gamma}_s = 0$. The measurements were performed with an ARES-G2 rheometer using an OWCh signal. For reference, the noise and inertia limits of the device are also included on the figure. The experimental results agree well with the expected viscosities. The intermediate frequency range ($1 - 10$ rad/s) is influenced by the natural frequency of the transducer $\omega_n = \sqrt{K/M}$, with M and K the combined transducer and geometry mass, and the instrument stiffness, respectively. The corresponding values are listed in Table 1 resulting in observable resonance peaks at $\omega_n = 2.45$ rad/s and $2\omega_n = 4.9$ rad/s. At high frequencies, above a frequency of 80 rad/s, a sudden broadening of the measured material property (here the magnitude of the dynamic viscosity $\eta^*_\perp(\omega)$) is observed. This rapid localized divergence, or "blowout" as it is henceforth referred to, cannot be explained by any material characteristic, nor by any of the mechanical limits of the rheometer, suggesting that it is a measurement artefact. Hudson-Kershaw et al. (2024) showed that a similar artefact is observed when the logarithmic skew rate of the waveform is defined with insufficient numerical precision, eliminating power present in the waveform at the highest frequencies and resulting in significant noise appearing in the

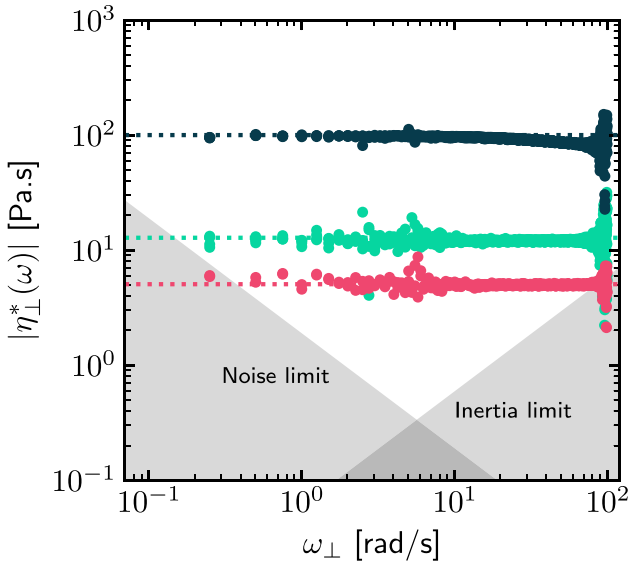


Fig. 1 Orthogonal frequency sweeps of Brookfield standards with viscosities of 100 Pa s (dark blue), 12.8 Pa s (green) and 5.05 Pa s (red) performed using an OWCh signal ($T = 25$ s, $\omega_1 = 0.1$ rad/s, $\omega_2 = 100$ rad/s, $r = 0.1$, $\gamma_0 = 4.85$ %). The dotted lines are the expected and verified shear viscosities. At high frequencies, the magnitude of the complex viscosity $|\eta^*_\perp(\omega)|$ begins to diverge creating a high-frequency "blowout". For clarity, the noise limit $|\eta^*_\perp(\omega)| = (K_{\perp,\sigma}F_{min})/(\omega\gamma)$ and the inertia limit $|\eta^*_\perp(\omega)| = K_{\perp,\sigma}/K_{\perp,\gamma}(M\omega)$ of the device are also depicted. Here, $K_{\perp,\sigma}$ and $K_{\perp,\gamma}$ are the stress and strain constants of the orthogonal double walled Couette geometry, F_{min} is the minimal transducer normal force limit and M is the combined transducer and geometry mass. The values for these constants are listed in Table 1

Table 1 ARES-G2 instrument parameters used in the calculation and correction of the measured moduli

Parameter	Symbol	Value	Unit
Transducer mass	M	0.662	kg
Friction coefficient	ζ	8.82	N m^{-1} s^{-1}
Spring constant	K	3.98	N m^{-1}
Orthogonal stress constant	$K_{\perp,\sigma}$	92.27	Pa N^{-1}
Orthogonal strain constant	$K_{\perp,\gamma}$	1081.03	m^{-1}
Orthogonal end-factor	$c_\perp$	0.776	[-]

measured material properties. However, this truncation effect was accounted for during the measurements presented here and should not contribute to the observed "blowout". Instead, because the chirp signal $\gamma_{owch}(t)$ is tapered at high frequencies by the presence of the Tukey window, the driving strain amplitude is progressively reduced towards the inherent instrument noise level, effectively reducing the signal-to-noise ratio (SNR) of the measurement at high frequencies. This example shows that a reduction in SNR creates a noise-dominated regime, resulting in the aforementioned broadening of the measured dynamic viscosity $\eta^*_\perp(\omega)$, or equivalently the complex modulus $G^*_\perp(\omega) = i\omega\eta^*_\perp(\omega)$ at high frequencies. The present paper aims to investigate the detailed origins of this artefact and then use this understanding to propose a modification to the chirp signal that reduces the impact of this high-frequency "blowout". Numerical simulations with a linear Maxwell model provide a verification of the proposed correction strategies. As a result, a novel double exponential chirp (DEC) signal is proposed that helps avoid the "blowout" and provide more accurate high-frequency measurements. Additional simulations using the nonlinear Giesekus model combined with verification experiments on a wormlike micellar fluid are used to define the working conditions that optimize this novel approach.

Materials and methods

The OrthoChirp simulations of the Maxwell fluid and the Giesekus model are discussed first. Subsequently, the synthesis of the wormlike micellar solution is summarized together with the in-house rheological setup.

Simulations with a linear viscoelastic Maxwell fluid

To understand the root cause of any high-frequency anomalies, first numerical simulations are performed using the linear Maxwell model equivalent to a Newtonian dashpot and a Hookean spring in series (Maxwell 1867; Macosko 1994; Morrison 2001):

$$\frac{d\sigma(t)}{dt} = G\dot{\gamma}(t) - \frac{\sigma(t)}{\tau} \tag{5}$$

As the high-frequency effects are anticipated from Fig. 1 to be material independent, the use of the Maxwell model as the simplest viscoelastic approximation is justified. Here, $\sigma(t)$ is the shear stress, $\dot{\gamma}(t) = d\gamma(t)/dt$ the applied shear rate, G the shear modulus and τ the relaxation time of the material. For illustrative purposes, G and τ are arbitrarily chosen to be 50 Pa and 1 s. The strain input $\gamma(t)$ is an OWCh signal $\gamma_{owch}(t)$ given by (3)–(4) with strain amplitude $\gamma_0 = 5$ %. A start frequency $\omega_1 = 0.1$ rad/s and end frequency $\omega_2 = 100$ rad/s are used ($W = 1000$). The duration of the chirp is set to the minimal required time $T_{min} = \lceil 2\pi/\omega_1 \rceil = 63$ s necessary to resolve the lowest frequency component of the chirp (ω_1) in the Fourier domain. Four different values of the Tukey parameter r are used to investigate high-frequency effects: 0, 0.1, 0.3 and 0.5. It should be noted that Geri et al. (2018) have shown that $r \in [0.06, 0.15]$ provides minimal error for the OWCh signal on typical rheometric instrumentation. The shear strain and shear rate are sampled at 1 kHz. Equation (5) is solved numerically for an input signal $\gamma_{owch}(t)$ using the explicit Runge–Kutta method of order 8 within the *solve_ivp()* function of the SciPy package (Virtanen et al. 2020). The complex modulus $G^*(\omega)$ of the material is calculated

directly from Fourier transforms of the input forcing and output signal by the following:

$$G^*(\omega) = \frac{\tilde{\sigma}(\omega)}{\tilde{\gamma}(\omega)} \tag{6}$$

where the tilde symbol ~ represents the Fourier transform of the signal. The Fourier transforms of the strain input $\gamma_{owch}(t)$ and stress output $\sigma_{owch}(t)$ signal are calculated using the FFT algorithm, provided by the same SciPy package (Virtanen et al. 2020). No zero-padding or other post-processing was performed prior to performing the Fourier transformation.

In order to investigate the effect of the signal-to-noise ratio, Gaussian-distributed noise $\Delta \sim N(\mu, \sigma^2)$, with mean μ and variance σ^2, is added to the input strain and the resulting stress response in the time domain:

$$\begin{aligned} \gamma_n(t) &= \gamma(t) + \Delta_\gamma(t) \quad ; \quad \Delta_\gamma = N(0, 0.01) \\ \sigma_n(t) &= \sigma(t) + \Delta_\sigma(t) \quad ; \quad \Delta_\sigma = N(0, 1) \end{aligned} \tag{7}$$

$$G_n^*(\omega) = G^*(\omega) + \Delta^*(\omega) = \frac{\tilde{\sigma}}{\tilde{\gamma}} + \frac{\tilde{\Delta}_\sigma \tilde{\gamma} - \tilde{\Delta}_\gamma \tilde{\sigma}}{\tilde{\gamma}(\tilde{\Delta}_\gamma + \tilde{\gamma})} \tag{8}$$

where $\Delta_\gamma(t)$ and $\Delta_\sigma(t)$ represent the noise vectors of the strain input and stress output, respectively. The noise vectors are considered uncorrelated since the actual applied strain and the resulting stress output are assumed to be measured independently on any separate motor-transducer (SMT) rheometric device. From initial analysis of the raw signals of the results in Fig. 1, it is apparent that the added noise on the output stress signal is significantly larger than the noise in the strain input signal. This observation is reflected in the larger standard deviation used for Δ_σ. Quantifying the amount of background noise present with respect to the material response is most conveniently done using the signal-to-noise ratio (SNR), typically reported in decibels, which is calculated from the ratio of the amplitudes of the signal over the noise (Haykin and Moher 2007; Semmlow 2012):

$$\text{SNR} = 20\log_{10}\left(\frac{|G^*(\omega)|}{|\Delta^*(\omega)|}\right) = 20\log_{10}\left(\frac{|\tilde{\sigma}(\tilde{\Delta}_\gamma + \tilde{\gamma})|}{|\tilde{\Delta}_\sigma \tilde{\gamma} - \tilde{\Delta}_\gamma \tilde{\sigma}|}\right) \tag{9}$$

A SNR $\gg 0$ indicates that the amplitude of the signal due to the material response, which is proportional to the signal power, is larger than the amplitude, or power, of the background noise. In this limit, the correct material properties can be computed from the noisy stress and strain signals. For a SNR < 0, the desired signal arising from the material response will be significantly affected by the background noise making accurate measurement of the frequency-dependent material properties difficult.

Simulations with the nonlinear Giesekus model

To validate the working of the new double exponential chirp and optimize the key parameters (explained in further detail in "Redistributing power: double exponential chirps" below), wormlike micellar (WLM) solutions, known to be accurately described by the Giesekus model (Giesekus 1982; Kim et al. 2013), are experimentally investigated with optimally windowed OrthoChirps in "Verification with wormlike micellar solutions". Numerical simulations utilizing the Giesekus model are used to determine the optimal range for the experimental input parameters.

The Giesekus model is described by Giesekus (1982); Kim et al. (2013):

$$\boldsymbol{\sigma} + \tau\overset{\triangledown}{\boldsymbol{\sigma}} + \frac{\alpha}{G}\boldsymbol{\sigma}\cdot\boldsymbol{\sigma} = \tau G\dot{\boldsymbol{\gamma}} \tag{10}$$

with $\boldsymbol{\sigma}$ the extra stress tensor, $\dot{\boldsymbol{\gamma}}$ the rate of deformation tensor, α the dimensionless mobility parameter and $\overset{\triangledown}{\boldsymbol{\sigma}}$ the upper-convected time derivative of the stress tensor:

$$\overset{\triangledown}{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} + \boldsymbol{v}\cdot\nabla\boldsymbol{\sigma} - \left[(\nabla\boldsymbol{v})^T\cdot\boldsymbol{\sigma} + \boldsymbol{\sigma}\cdot\nabla\boldsymbol{v}\right] \tag{11}$$

For orthogonal superposition of an oscillatory perturbation $\dot{\gamma}_\perp$, in this case a chirp signal, on a steady rotational shear flow $\dot{\gamma}_s$, the velocity vector $\boldsymbol{v}$ can be written as follows:

$$\boldsymbol{v} = \left[\dot{\gamma}_s x_2,\ 0,\ \dot{\gamma}_\perp x_2\right]^T \tag{12}$$

Combining (10)–(12) results in six coupled ordinary differential equations:

$$\sigma_{11} - \tau(2\dot{\gamma}_s\sigma_{12} - \dot{\sigma}_{11}) + \frac{\alpha}{G}(\sigma_{11}^2 + \sigma_{12}^2 + \sigma_{13}^2) = 0 \tag{13}$$

$$\sigma_{12} - \tau(\dot{\gamma}_s\sigma_{22} - \dot{\sigma}_{12}) + \frac{\alpha}{G}(\sigma_{11}\sigma_{12} + \sigma_{12}\sigma_{22} + \sigma_{13}\sigma_{23}) = G\dot{\gamma}_s\tau \tag{14}$$

$$\sigma_{13} - \tau(\dot{\gamma}_\perp\sigma_{12} + \dot{\gamma}_s\sigma_{23} - \dot{\sigma}_{13}) + \frac{\alpha}{G}(\sigma_{11}\sigma_{13} + \sigma_{12}\sigma_{23} + \sigma_{13}\sigma_{33}) = 0 \tag{15}$$

$$\sigma_{22} + \tau\dot{\sigma}_{22} + \frac{\alpha}{G}(\sigma_{12}^2 + \sigma_{22}^2 + \sigma_{23}^2) = 0 \tag{16}$$

$$\sigma_{23} - \tau(\dot{\gamma}_\perp\sigma_{22} - \dot{\sigma}_{23}) + \frac{\alpha}{G}(\sigma_{12}\sigma_{13} + \sigma_{22}\sigma_{23} + \sigma_{23}\sigma_{33}) = G\dot{\gamma}_\perp\tau \tag{17}$$

$$\sigma_{33} - \tau(2\dot{\gamma}_\perp\sigma_{23} - \dot{\sigma}_{33}) + \frac{\alpha}{G}(\sigma_{13}^2 + \sigma_{23}^2 + \sigma_{33}^2) = 0 \tag{18}$$

An analytical solution for the stress tensor in (13)–(18) is provided by Kim et al. (2013) with the constraint that the oscillatory orthogonal perturbation is small compared to the steady shear flow, $\dot{\gamma}_\perp/\dot{\gamma}_s \ll 1$, such that σ_{13} and σ_{23} can be ignored in (13)–(15). The physical interpretation of this

simplification is that the steady shear flow is not affected by the small amplitude of the oscillatory perturbation, while the orthogonal perturbation can be influenced by the strong steady shear.

For the simulations, the parameters are set in accordance with the results obtained by Kim et al. (2013): $G = 37$ Pa, $\tau = 1.5$ s, $\eta_s = 36$ mPa s and $\alpha = 0.5$. The solvent viscosity η_s is accounted for by an additional solvent contribution to the stress $\boldsymbol{\sigma}_s = \eta_s \dot{\boldsymbol{\gamma}}$ resulting in the linear addition of an additional term $\eta_s \omega$ in $G''(\omega)$ and an extra term $\eta_s \dot{\gamma}_s$ in the steady shear rate. The orthogonal strain input $\gamma_\perp(t)$ is either an OWCh signal $\gamma_{owch}(t)$ or a double exponential chirp (DEC) $\gamma_{dec}(t)$ as further described in "Redistributing power: double exponential chirps". The DEC has two important parameters, a and b, fixing the inflection point within the time-frequency spectrum. During the optimization process, a and b are varied from 0.5 to 1.0. A grid of possible combinations are simulated to provide a contour plot of the corresponding SNR. The strain amplitude $\gamma_0 = 4.85\,\%$ corresponding to a 45 μm displacement on an ARES-G2 rheometer, which is also used for all experiments. The duration of each chirp is $T = 65$ s and is sampled at 1 kHz. Similar to the simulations with the Maxwell model, the noise added to these simulations again follows a Gaussian distribution but the average value and standard deviation are modified to represent the values of the experimentally determined noise on the actual rheometer as closely as possible:

$$\begin{aligned} \Delta_\gamma &= N(0,\ 0.001) \\ \Delta_\sigma &= N(0.05,\ 0.008) \end{aligned} \tag{19}$$

The equations are solved numerically using the same explicit Runge–Kutta method of order 8 as the simulations with the Maxwell model. The resulting dynamic moduli and the signal-to-noise ratio SNR are calculated according to (6)–(9) using the σ_{23} component of the stress tensor.

To generalize the optimization of a and b to materials that have different characteristics compared to the wormlike micellar fluid, a separate set of simulations are performed in which G and τ are varied from [0.1, 1, 10, 100, 1000, 10000] Pa and [0.063, 0.353, 1.987, 11.173, 62.832] s, respectively. The values of τ are such that the critical frequencies $\omega_c = 2\pi/\tau$ are logarithmically divided over the range of measured frequencies measured by the chirp (0.1–100 rad/s).

Rheological verification with wormlike micellar solutions

Following the aforementioned simulations, orthogonal frequency sweeps are performed on wormlike micellar (WLM) solutions. The WLM solution consists of 100 mM sodium chloride (NaCl; CAS: 7647-14-5), 100 mM cetylpyridinium chloride (CPyCl; CAS: 6004-24-6) and 50 mM sodium salicylate (NaSal; CAS: 54-21-7). An aqueous solution of NaCl and NaSal is prepared first in a volumetric flask. After full dissolution, the CPyCl is added to the solution. The final solution is magnetically stirred for 24 h to obtain a homogeneous solution and to remove entrapped air bubbles.

The measurements are performed on a TA Instruments strain-controlled ARES-G2 rheometer with a double wall concentric cylinder geometry, modified with slots at the bottom of the inner cylinder and slots at the top of the bob. The Couette cell has an inner and outer diameter of 27.7 mm and 34.0 mm, respectively, with the inner cylinder height equaling 51.6 mm. The bob has an inner and outer diameter of 29.4 mm and 32.1 mm, respectively. The slots at the top mitigate surface tension effects at the free surface while those at the bottom minimize the effect of pressure-driven oscillatory pumping flows (Franck 2013). The measurement gap between the bottom of the bob and the cup is 8 mm to avoid obstruction of the bottom slots of the cup by the bob. To ensure that the loading protocol is identical between experiments, the upper bob geometry is lowered with a controlled rate of 0.1 mm/s after the sample is loaded into the lower cup. The sample is pre-sheared at 1 s^{-1} for 600 s followed by a waiting period of 60 s for stress relaxation. Due to the short duration of the chirp signal (on the order of $T_{min} = 60$ s), and the minimal contact area with air, sample evaporation is assumed to be negligible during the experiments.

Steady shear rates of $\dot{\gamma}_s = 0$ s^{-1}, 1 s^{-1} and 10 s^{-1} were applied through the TRIOS software of the rheometer. The orthogonal chirp signals are not applied using the TRIOS software but instead introduced externally via a custom setup for enhanced control without the need for any hardware or firmware modifications to the rheometer. As initially described by Kim et al. (2013), an oscillatory voltage signal, representing the desired displacement of the transducer, is generated externally and sent to the *osp-in* port of the rheometer by an acquisition board, interfaced with a NI PCI-6281 analog-to-digital converter through LabVIEW (NI Instruments). The actual displacement profile $z(t)$ and corresponding normal force profile $F(t)$ are measured by acquiring the voltages coming out of the *osp-out* and *normal-out* ports with the same acquisition board. The raw signals as a function of time are converted from volts to their physical displacement or force units according to calibrated conversion factors. The measured constant normal force offset that arises from the weight of the suspended bob geometry is removed by subtracting the average value of the signal. The resulting signals are subsequently Fourier-transformed to obtain the corresponding frequency domain signals $\tilde{z}(\omega)$ and $\tilde{F}(\omega)$. The desired displacement sent to the *osp-in* port is actually a reference profile $z_{ref}(t)$ for the ARES-G2 force-rebalance transducer (Franck 2013). Internally, the normal

force control loop generates a driving force based on the difference between the reference signal and the actual measured profile $z(t) - z_{ref}(t)$, similar to a stress-controlled rheometer performing a strain-controlled experiment. Thus, since the orthogonal superposition of an oscillatory strain applied through the normal force transducer operates similarly to a stress-controlled rheometer, it is also required to correct for instrument inertia and compliance, and frictional losses in the air bearing. The corrected complex modulus is calculated following Vermant et al. (1997):

$$G^*_{corr}(\omega) = \frac{c_\perp K_{\perp,\sigma}}{K_{\perp,\gamma}} \left[\frac{\tilde{F}(\omega)}{\tilde{z}(\omega)} - (K - M\omega^2) - i\zeta\omega \right] \tag{20}$$

where M represents the mass of the system, ζ the friction coefficient and K the spring constant of the force-rebalance transducer in the axial direction. The orthogonal stress and strain constant $K_{\perp,\gamma}$ and $K_{\perp,\gamma}$ are solely dependent on the geometry, and $c_\perp$ is the orthogonal end-factor that accounts for any residual surface tension and pressure flow effects (Tao and Forster 2020a, b). The values for all parameters can be found in Table 1 as determined by calibration beforehand. Finally, the storage modulus $G'(\omega)$ and loss modulus $G''(\omega)$ can be extracted as the real and imaginary part of the complex signal $G^*_{corr}(\omega)$, respectively.

Results and discussion

Simulations of the Maxwell model: decreasing SNR due to windowing

To gain a qualitative understanding of the cause of the signal divergence or "blowout" when performing experiments, the dynamic moduli and SNR measured from the chirp pulse are calculated for the linear Maxwell model from (6)–(9) and are depicted in Fig. 2. From Fig. 2a, it is apparent that the results for all simulations with different tapering parameter r coincide for the lower frequencies and agree with the analytical solution of the Maxwell model, shown as the dotted line. However, at high frequencies, the "blowout" can be observed when the Tukey window in (4) is applied to the chirp signal ($r \neq 0$). The severity of the "blowout" increases with increasing tapering severity and the onset shifts toward lower frequencies. Figure 2b shows the corresponding SNR of the simulation results calculated by (9). The occurrence of the "blowout" appears to be correlated with the onset of a decreasing SNR. The critical SNR, defined as the value at which the ratio $G^*(\omega, r)/G^*(\omega, 0)$ deviates by more than 10 % ($G^*(\omega, r)/G^*(\omega, 0) \leq 0.9$), is approximately 15 dB for all simulated values of r. At the highest frequencies, the SNR becomes negative, indicating that the contribution of

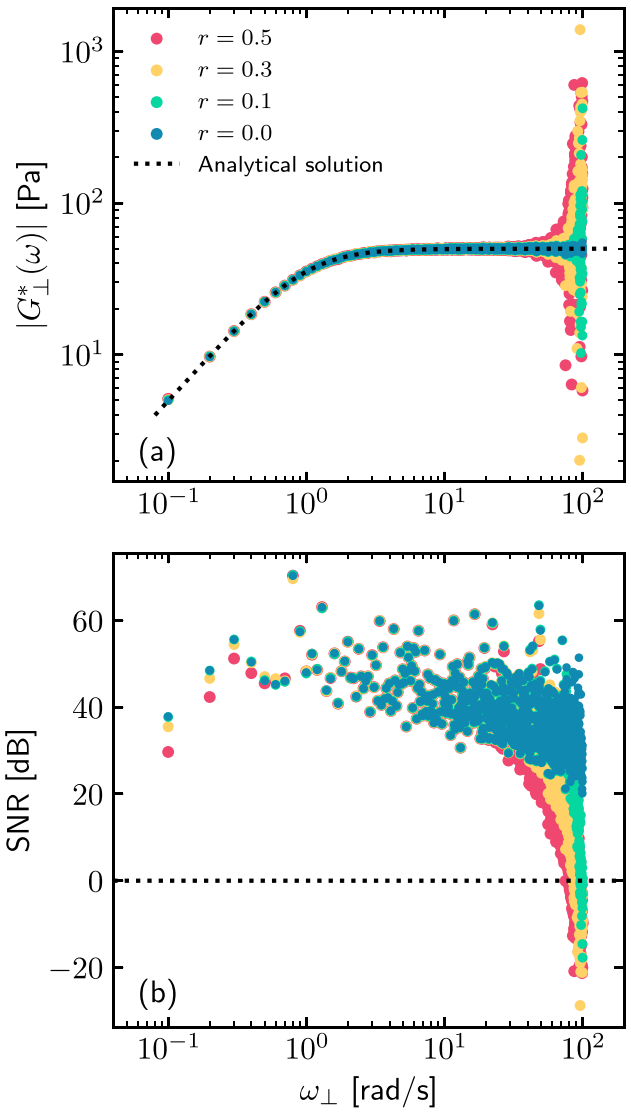


Fig. 2 Numerical simulation results for the linear Maxwell model of (5) ($G = 50$ Pa, $\tau = 1$ s) for an OWCh input signal as given in (3)–(4) ($\gamma_0 = 4.85$ %, $\omega_1 = 0.1$ rad/s, $\omega_2 = 100$ rad/s, $T = 63$ s) for different values of the tapering parameter r. **a** The magnitude of the complex modulus as a function of frequency from (8). **b** The accompanying SNR ratios for the different r, calculated using (9). The "blowout" appears at high frequencies with the addition of a tapering window. The severity of the "blowout" is proportional to the tapering parameter r and coincides with the onset of a decrease in the SNR

the noise towards the total power of the measured signal is larger than the contribution arising from the material response. The addition of a tapering window that modulates the amplitude of the driving chirp signal creates a noise-dominated regime at high frequencies.

Although the tapering window is symmetric, the same "blowout" effect is not observed at low frequencies. The frequency spectrum of a chirp signal is inversely proportional to the square root of the frequency ($\sim\omega^{-1/2}$), known

as a pink frequency spectrum (Rathinaraj et al. 2022). The power spectral densities (PSD) $|\tilde{\sigma}(\omega)|^2$ of the simulated shear stresses of the results in Fig. 2 are shown in Fig. 3, where this pink trend can be observed. The signal power contained within the low frequencies is higher than that contained within the high frequencies, making material property determination more robust against the effects of noise. As the tapering (r) increases, the PSD at high frequencies is significantly reduced, making the results progressively more prone to the effects of high-frequency noise.

Redistributing power: double exponential chirps

To reduce the high-frequency "blowout" caused by the rapidly decreasing SNR induced by the presence of the tapering window, a redistribution of the signal power across the frequency range is required. Every chirp signal can be described by the general form $x(t) = x_0 \sin[\phi(t)]$ with $\phi(t)$ the instantaneous phase of the signal, which is related to the instantaneous frequency $\omega(t)$ by integration over time, i.e., $\omega(t) = d\phi(t)/dt$. Since the complex modulus $G^*(\omega)$ computed from (6) is related to the frequency spectra of both the input and output chirp waveform, the current work takes the approach of designing a new instantaneous frequency function, thereby reshaping the PSD of the signal, and integrating the result to obtain a new chirp signal, reducing the effect of the "blowout".

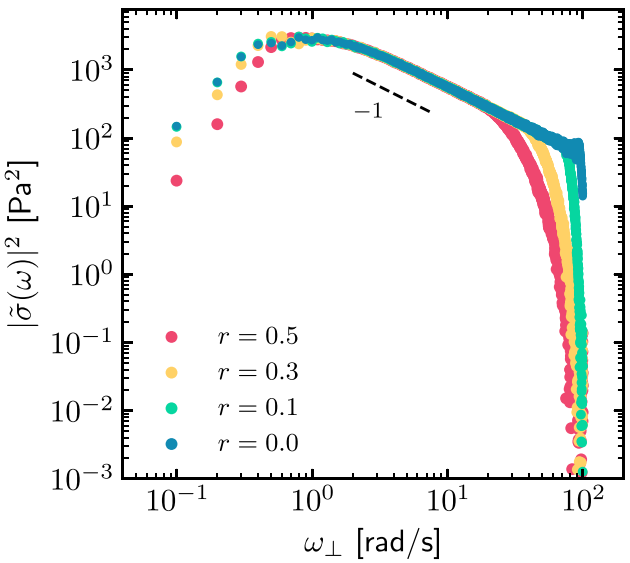


Fig. 3 Power spectral densities (PSD) of the computed shear stress response. The PSD can be characterized as a pink frequency spectrum. Material properties computed at lower frequencies are less affected by noise due to the increased power in the signal here. The effect of the Tukey window parameter r on the spectrum can also be observed at high frequencies

In order to redistribute the power towards higher frequencies, increasing the SNR in the Fourier domain, instead of the exponential spectrum of the OWCh signal, the use of an S-shaped curve is examined. The well-known sigmoid function was considered, but unlike the OWCh, the logarithmic slew rate $\alpha_{\ln}(t) = d\ln[\omega(t)]/dt$ of the sigmoid is no longer constant with time. Since the frequencies of interest are often logarithmically spaced, a constant logarithmic slew rate ensures constant time spent per frequency decade. In order to retain a constant logarithmic slew rate, and thus the exponential nature of the spectrum, instead two exponential functions are spliced in a piecewise fashion, similar to the bilateral or double exponential distribution encountered in statistics (Guthrie 2020). The instantaneous frequency of the new waveform is given by the following:

$$\omega(t) = \begin{cases} \omega_1 W^{t/(AT)} & \text{if } t \le aT \\ \omega_1 \left[DW - W^{(C-t)/(BT)}\right] & \text{if } t > aT \end{cases} \tag{21}$$

with $W = \omega_2/\omega_1$ and ω_1, ω_2 being the start and end frequency, respectively. The inflection point at which the two exponential functions intersect depends on the two user inputs a and b. The parameter $a \in (0, 1)$ fixes the locus of the inflection point in the time domain as a fraction of T. Similarly, the input parameter $b \in (1/W, 1)$ fixes the inflection point in the frequency domain as a fraction of the final frequency ω_2.

The parameters A, B, C and D in (21) are dependent on a and b and are calculated by asserting a set of conditions. The first condition $\omega(aT) = b\omega_2$ mathematically defines the inflection point and enables the determination of A when substituted in (21):

$$A = \frac{a\ln(W)}{\ln(bW)} \tag{22}$$

The second condition requires the instantaneous frequency $\omega(t)$ to be C^1-continuous, ensuring the continuity of both the phase $\phi(t)$ and the slew rate $\alpha(t) = d\omega(t)/dt$. This continuity is crucial in mechanical systems to prevent potential damage due to abrupt changes in acceleration. To guarantee C^1-continuity at the inflection point, the left- and right-sided limits of $\omega(t)$, denoted as $\omega(aT^-)$ and $\omega(aT^+)$, and their derivatives $\alpha(aT^-)$ and $\alpha(aT^+)$ must be equal. Applying these conditions leads to the following equations:

$$-DW + W^{a/A} + W^{(C-aT)/(BT)} = 0 \tag{23}$$

$$BW^{a/A} - AW^{(C-aT)/(BT)} = 0 \tag{24}$$

Finally, ensuring that the entire frequency range is covered regardless of the chosen user inputs, the condition $\omega(T) = \omega_2$ must be satisfied. This constraint allows the determination of

the unknown parameter D as a function of two other parameters B and C:

$$D = W^{(C-T)/(BT)-1} + 1 \tag{25}$$

Rearranging (24), substituting the input variable a from (22) and combining the result with (23) and (25), expressions for the parameters B and C are found:

$$b\left(1 + \frac{B}{A}\right) = 1 + W^x \tag{26}$$

$$C = \frac{T}{\ln(W)}\left[\ln(bW)(A+B) - B\ln\left(\frac{A}{B}\right)\right] \tag{27}$$

with

$$x = \frac{\ln(bW)(A+B) - B\ln(A/B) - \ln(W)}{B\ln(W)} - 1 \tag{28}$$

The expression for B in (26) is implicit and needs to be solved numerically. However, if the second term on the right-hand side becomes significantly smaller than 1, (26) simplifies allowing B to be solved for directly:

$$B \approx \left(\frac{1-b}{b}\right)A \tag{29}$$

To determine the conditions for which the above equation is valid, a permissible error $\varepsilon \ll 1$ is introduced in (26) such that

$$\left(\frac{\ln(bW)(A+B) - B\ln(A/B) - \ln(W)}{B\ln(W)} - 1\right)\ln(W) = \ln(\varepsilon), \tag{30}$$

which, after substitution of B using (29) and rearranging, yields a critical condition for a:

$$a_{crit} = \frac{b\ln(bW)}{b\ln(bW) + (1-b)\ln\left(\frac{1-b}{\varepsilon}\right)} \tag{31}$$

The expression for a_{crit} revolves around the choice of ε where a value of $\varepsilon = 1 \cdot 10^{-6}$ is used for further calculations. Finally, (21) is integrated with respect to time to determine the phase of the new double exponential chirp signal $\phi_{dec}(t)$:

$$\phi_{dec}(t) = \begin{cases} \frac{\omega_1 AT}{\ln(W)}\left[W^{t/(AT)} - 1\right] & \text{if } t \le aT \\ \frac{\omega_1}{\ln(W)}\left[AT\left(W^{a/A} - 1\right) + \right. & \\ \left. BTW^{C/BT}\left(W^{-t/(BT)} - W^{-a/B}\right) + DW\ln(W)(t - aT)\right] & \text{if } t > aT \end{cases} \tag{32}$$

Figure 4a compares the instantaneous frequencies of the new double exponential chirp (DEC) signal in (21) against the OWCh signal ($a = b = 1$ in (21)), highlighting the inflection points for clarity. A key advantage of the DEC signal is that the inflection point, set by the choice of the input variables a and b, can be adjusted in both the time and frequency domains providing an additional degree of freedom. This key advantage is crucial for the optimization discussed in "Optimization of input parameters through the Giesekus model". The DEC strain input $\gamma_{dec}(t) = \gamma_0 \sin[\phi_{dec}(t)]$ in Fig. 4b corresponds to the solid line in Fig. 4a. Windowing of the DEC signal is still required to avoid spectral leakage, similar to the original OWCh signal (Geri et al. 2018). Simulation results for the Maxwell model given in (5) with the DEC waveform given in (32) are shown in Fig. 4b and are compared to the previously obtained results with the

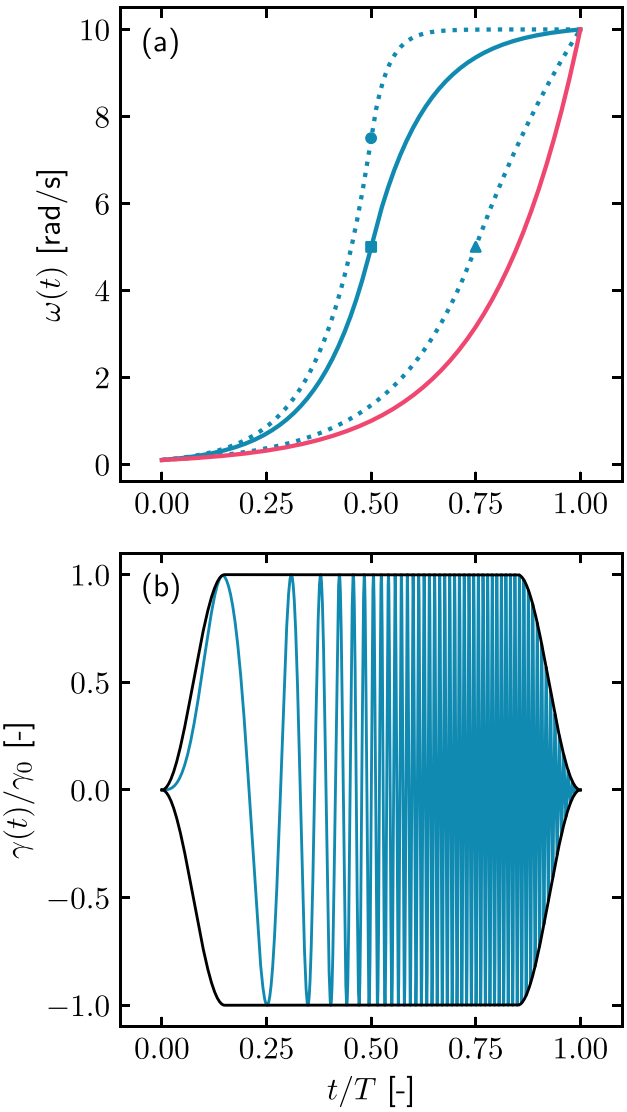


Fig. 4 **a** Instantaneous frequency $\omega(t)$ of three different DEC waveforms (blue) and an OWCh waveform (red) for $\omega_1 = 0.1$ rad/s, $\omega_2 = 10$ rad/s and a chirp duration of $T = 60$ s. Square: $(a, b) = (0.5, 0.5)$, Triangle: (0.75, 0.5). Circle: (0.5, 0.75); **b** the double exponential chirp (DEC) signal corresponding to the solid line in **a** $(a, b) = (0.5, 0.5)$ with a Tukey window ($r = 0.3$)

OWCh signal in Fig. 5 for a Tukey parameter $r = 0.3$. The simulation results show that the redistribution of the input signal power as a result of the new instantaneous frequency function diminishes the high-frequency "blowout".

Optimization of input parameters through the Giesekus model

The benefit of the DEC waveform over the OWCh depends on the values of the input parameters a and b, for which the optimal choice is not immediately obvious. To determine an optimal working window such that the SNR at the final chirp frequency ω_2 is maximized, the Giesekus model, because of its broader applicability, was solved numerically for various combinations of (a, b). The resulting contour plots of the SNR at $\omega_2 = 100$ rad/s are shown in Fig. 6 for $r = 0.1$ and $r = 0.5$. A critical SNR threshold of 15 dB, below which a "blowout" occurs, was identified from the Maxwell simulations in "Simulations of the Maxwell model: decreasing SNR due to windowing". Comparing Fig. 6a and 6b, the SNR generally increases diagonally over the parameter space (a, b). The lowest SNR is found for $a \approx 1$ and $b \approx 0.5$ in the lower right region of the figure, while the highest SNR is found for $a \approx 0.5$ and $b \approx 1$ in the upper left region. Since a high SNR is desirable, the latter region is considered optimal. The OWCh signal, which suffers from the high-frequency "blowout", corresponds to $b = 1$ and exhibits significantly lower SNR than the optimal DEC values,

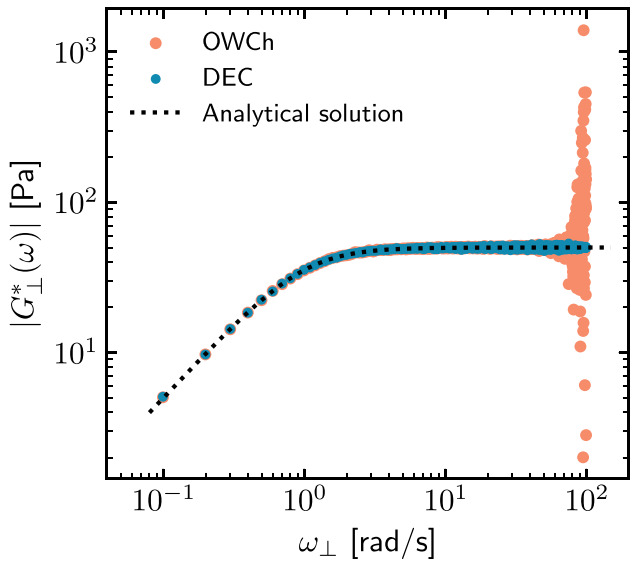


Fig. 5 Comparison of the numerical simulations results for a linear Maxwell fluid ($G = 50$ Pa, $\tau = 1$ s) for an OWCh input signal in (3) and a DEC input signal $\gamma_{dec}(t) = \gamma_0 \sin[\phi_{dec}(t)]$ with $\phi_{dec}(t)$ given by (32) ($\omega_1 = 0.1$ rad/s, $\omega_2 = 100$ rad/s, $\gamma_0 = 4.85$ %, $T = 65$ s, $r = 0.3$). The result for the OWCh is identical to Fig. 2a. For the DEC, $a = b = 0.5$. The redistribution of power in the frequency domain mitigates the high-frequency "blowout"

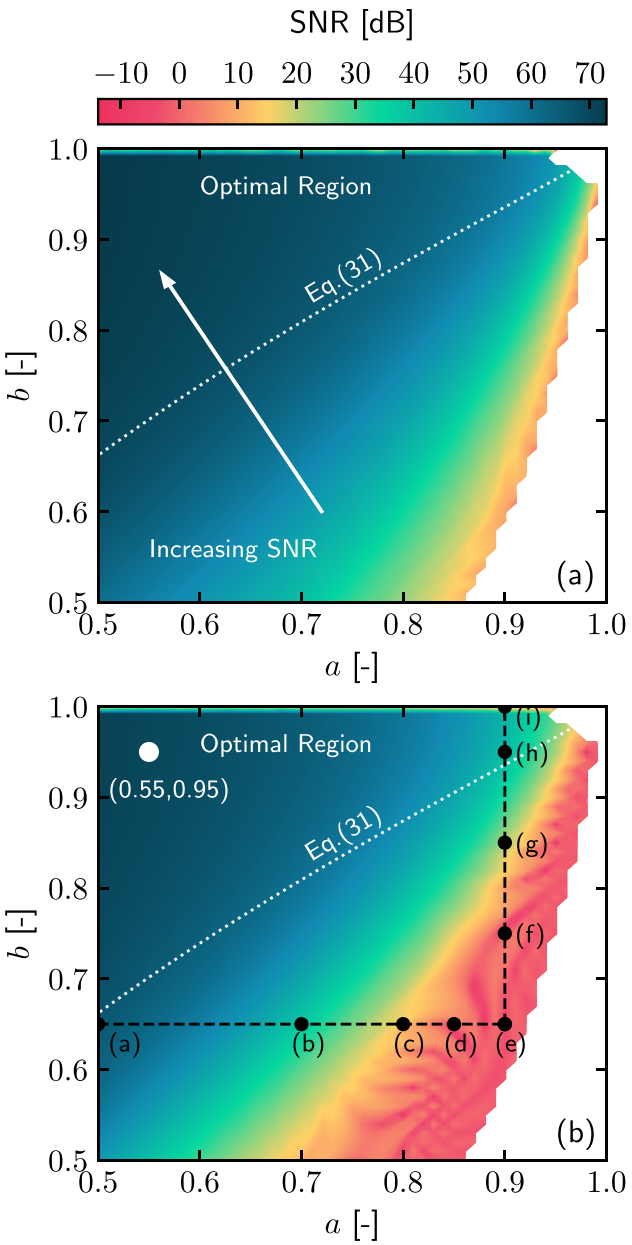


Fig. 6 Contour plots showing the SNR, as calculated by the Giesekus model with $G = 37$ Pa, $\tau = 1.5$ s, $\alpha = 0.5$, at $\omega = 100$ rad/s for different values of a and b. **a** $r = 0.1$ (b) $r = 0.5$. The SNR decreases diagonally from upper left to lower right. For certain combinations of (a, b), no DEC can be calculated as shown by the whitespace. The points numbered (a-i) and the point (0.55, 0.95) in **b** coincide with the values of (a, b) for the measurements performed on wormlike micellar solutions

highlighting the benefit of the new signal. For every value of the parameter b, there exists a maximum a-value beyond which the DEC waveform ceases to exist, and no solution is found for the Giesekus model. The expression for B in (26) no longer yields a positive and real solution, preventing the design of a proper DEC waveform beyond this limit. With

increasing Tukey parameter r (compare Fig. 6a to b), the onset of the tapering in the frequency domain shifts towards lower frequencies, as illustrated in Fig. 3. Consequently, the low-SNR region in Fig. 6 increases significantly. The values for a_{crit}, computed using (31), are also indicated on Fig. 6. The optimal SNR region can be found above this threshold, indicating that the approximation in (29) is valid for these choices of (a, b). Thus, $a < a_{crit}$ can be used as a sufficient but not necessary condition for identifying the optimal region.

The location of the optimal SNR region is not surprising. As the value of a decreases, the inflection point in Fig. 4 shifts to an earlier time, causing the time-varying signal to spend more time at high frequencies rather than at low frequencies, increasing the PSD at the high frequencies and reducing the effect of the noise. Essentially, the parameter a controls the distribution of signal power between low and high frequencies. On the other hand, the value of b controls the distribution of signal power across the high frequencies. As b approaches 1, the instantaneous frequency $\omega(t)$ flattens at high frequencies, as illustrated in Fig. 4 by comparing the S-curve for $(a, b) = (0.5, 0.5)$ and $(0.5, 0.75)$. This flattening allows the highest frequency component (typically 100 rad/s) to persist for a longer duration compared to the other high-frequency components, leading to an increased PSD.

To further understand the underlying mechanism, the SNR was analyzed as a function of the slew rate $\alpha(t)$ as defined in "Redistributing power: double exponential chirps" and evaluated at the highest frequency ω_2, which equates to the slew rate evaluated at the end of the imposed chirp signal, $\alpha(T)$. The slew rate for the DEC waveform with instantaneous frequency $\omega_{dec}(t)$ given by (21) is as follows:

$$\alpha_{dec}(T) = \left.\frac{d\omega_{dec}(t)}{dt}\right|_{t=T} = \frac{\omega_2 \ln(W)}{BT}(D-1) \tag{33}$$

The slew rate $\alpha_{dec}(T)$ depends on both input parameters a and b through the implicit dependence of B and D on the coefficients, allowing the contour plots in Fig. 6 to collapse into a single scatter plot. The evolution in the resulting SNR, calculated using (9), as a function of the slew rate $\alpha(T)$ is replace below in Fig. 10a. The results for the simulations show that increasing the slew rate initially causes a sharp decrease in SNR, after which the ratio stabilizes at a plateau value, which is dependent on the inherent noise floor of the rheometer. As $\omega_{dec}(t)$ flattens, the slew rate $\alpha_{dec}(T)$ approaches zero. However, according to (33), $\alpha_{dec}(T)$ only reaches zero when $D = 1$. From the definition of the parameter D in (25), this would require the exponent to approach $-\infty$, meaning that in practice, the slew rate $\alpha_{dec}(T)$ can never be zero (expect in the case of traditional discrete SAOS tests at a single frequency) but only asymptotically approaches zero. The results of the simulations in Fig. 10 provide insight into the influence of the Tukey parameter r. As the tapering severity increases, the SNR becomes more sensitive to changes of the slew rate.

Further insight into the optimal region for materials with different viscosities (different G and τ) is obtained by performing additional simulations as outlined in "Materials and methods". The resulting contour plots (data not shown) remain similar to Fig. 6, and the optimal region consistently occurs at low a and high b, confirming our initial findings. Instead, only the critical SNR contour line of 20 dB is presented in Fig. 7. The position of the critical contour line is directly related to the extent of the optimal region under the influence of the shear modulus G of the material. As the influence of noise becomes progressively smaller, the whole plane of the contour plot will shift to higher SNR and the 20 dB line is expected to move to higher values of a. As G increases, the critical SNR contour indeed shifts towards higher values of a, indicating an expanding optimal region. The applied strain results in a larger shear stress response for increasing shear modulus G, causing the shear stress to be less affected by transducer noise, leading to an improved SNR across the parameter space (a, b). No difference in optimal region is observed once the moduli become larger than 10 Pa. Notably, varying the relaxation time τ has no effect on the SNR. The small oscillations observed on the different 20 dB lines are attributed to the limited resolution of the simulations and are considered artefacts.

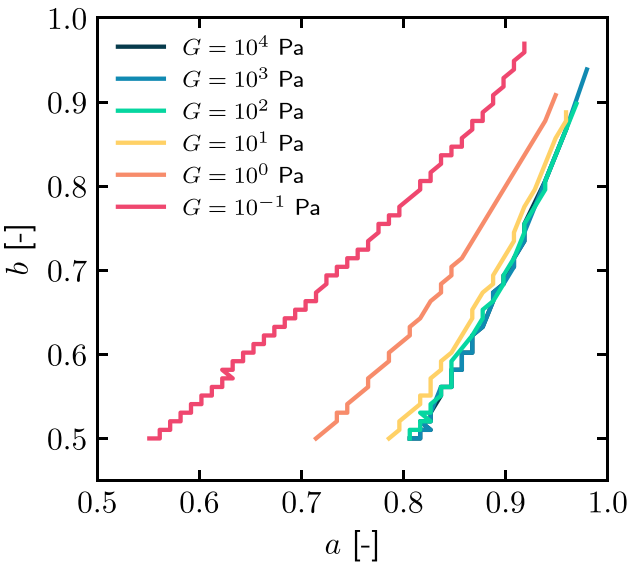


Fig. 7 Evolution of the 20 dB SNR line for different values of the shear modulus G. The 20 dB line shifts towards the right for higher values of the shear modulus, indicating that the optimal region (above the line) enlarges and noise has progressively less effect. No significant difference is observed once the moduli become larger than approximately 10 Pa

Verification with wormlike micellar solutions

To validate the simulation results from the previous section and experimentally test the performance of the DEC waveform, orthochirp measurements were conducted on a wormlike micellar (WLM) solution. The concentration of the WLM solution was selected such that the shear modulus $G \approx 30$ Pa is close to the minimal resolvable moduli limit of the rheometer with the orthogonal double walled Couette cell ($G_{min} \approx 2$ Pa). This ensures that noise has an observable impact on the measurements, allowing validation of the workings of the DEC waveform and comparison with the original OWCh signal.

First, measurements were conducted following the optimization results of "Optimization of input parameters through the Giesekus model" to validate their predictions. Multiple sets of the input parameters (a, b) were tested corresponding to points a–i in Fig. 6b, with the corresponding results presented in Fig. 8. The experimental data conforms well to the Giesekus model using parameters $G = 31.80$ Pa, $\tau = 0.684$ s and $\eta_s = 0.125$ Pa s. As predicted by the simulations, increasing a from 0.50 to 0.90 leads to a more pronounced "blowout" effect, while increasing b at constant a reduces the severity of the "blowout". Notably, the measurements in Fig. 8a and b yield good results despite the condition in (31) not being satisfied, confirming that

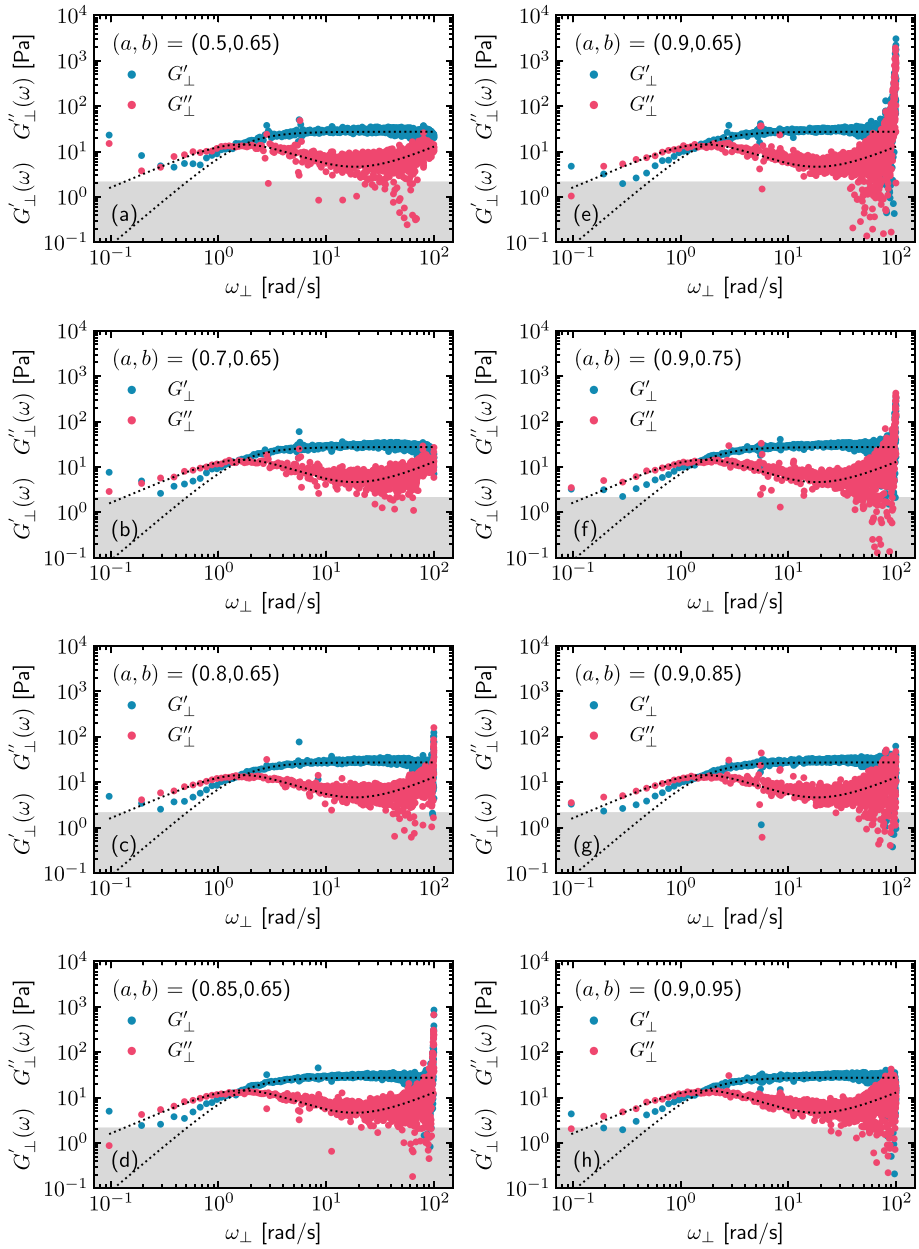


Fig. 8 OrthoDEC measurements on wormlike micellar solutions for different values of (a, b), as described in Fig. 6 ($r = 0.5$, $\dot{\gamma}_s = 0$ s^{-1}, $\gamma_\perp = 4.85\%$). Increasing the values of a increases the magnitude of the "blowout" as shown in panels (1–4) while increasing b decreases the "blowout" effect as can be seen in panels (5-8). The dotted lines are the analytical solution to the Giesekus model (Kim et al. 2013) using $G = 31.80$ Pa, $\tau = 0.684$ s and $\eta_s = 0.125$ Pa s

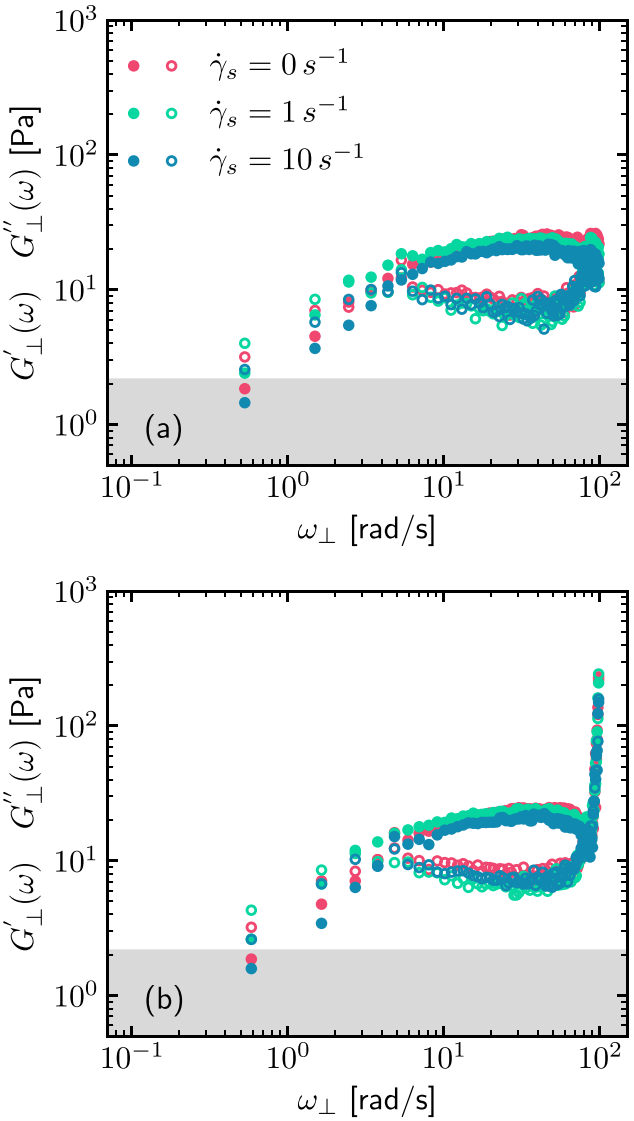


Fig. 9 Comparison of the orthogonal storage (filled) and loss (open) moduli for the DEC (**a**) and the OWCh (**b**) protocol for WLM solutions under different steady shear rates $\dot{\gamma}_s$ (0, 1, 10 s^{-1}). For the DEC and the OWCh, $(a, b) = (0.55, 0.95)$ and $(a, b) = (0.90, 1)$, respectively

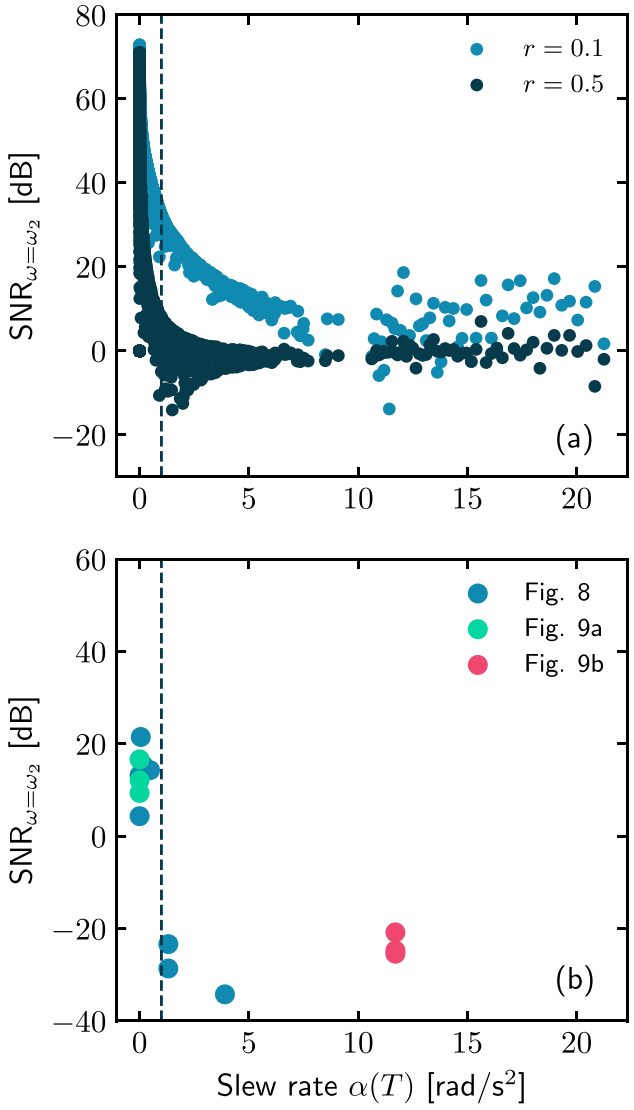


Fig. 10 Signal-to-noise ratio as a function of the slew rate $\alpha(t)$ evaluated at $t = T$ corresponding to the maximum frequency $\omega = \omega_2$. **a** For the simulation results of Fig. 6, the SNR decreases sharply with increasing slew rate, after which a plateau value is reached. For a Tukey parameter $r = 0.5$, the decrease at high values of $\alpha(T)$ is more severe than that observed for $r = 0.1$. **b** For the experimental results of Fig. 8 and Fig. 9, a similar trend is observed. The maximum recommended slew rate $\alpha_{max}(T) = 1$ rad/s^2 is represented by the dotted line

the criterion $a \leq a_{crit}$ is indeed a sufficient but not necessary condition.

Second, the DEC waveform is compared to the OWCh protocol for orthogonal superposition measurements with different shear rates. For the DEC protocol, $(a, b) = (0.55, 0.95)$ provides the best SNR according to the simulations. The OWCh protocol used is represented by point (i) in Fig. 6 with $(a, b) = (0.90, 1)$. The frequency resolution of the Fourier transform differs between the DEC and the OWCh waveform since the OWCh does not exist for $t > aT$ (aT is considered to be the total signal duration) such that the resolution in the frequency domain is $\Delta f = 2\pi/aT$. In contrast, for the DEC waveform, the frequency resolution is simply given by $\Delta f = 2\pi/T$ (T is the total signal duration of the DEC). This difference will not affect the results at the highest frequencies of interest here. To further increase the SNR, the results presented were boxcar-averaged over 10 points (Van Dusschoten and Wilhelm 2001). The results, plotted in Fig. 9, demonstrate that the DEC signal outperforms the OWCh signal, particularly at high frequencies. The "blowout" effect is significantly reduced, as evident when comparing Fig. 9a

with b. Additionally, with increasing shear rate, the values of both the storage and loss moduli drop, consistent with the predictions made by Kim et al. (2013).

Similar to the simulation results, the SNR was again analyzed as a function of the slew rate $\alpha(T)$ at the highest frequency. For the experimental measurements, the SNR was calculated following (9):

$$\mathrm{SNR} = 20\log_{10}\left(\frac{|G^*(\omega)|}{|\Delta^*(\omega)|}\right) = 20\log_{10}\left(\frac{|G_g^*(\omega)|}{|G_n^*(\omega) - G_g^*(\omega)|}\right) \tag{34}$$

where $G_n^*(\omega)$ are the noisy experimentally measured moduli and $G_g^*(\omega)$ are the noise-free moduli obtained from the Giesekus model fit. The evolutions in the resulting SNR for the measurement data are shown as a function of the slew rate $\alpha(T)$ in Fig. 10b and exhibit the same qualitative behaviour as the simulations, essentially a rapid improvement in the SNR as the slew rate is reduced. Based on the combined simulation and measurement observations, a maximum slew rate of $\alpha_{max}(T) < 1\,\mathrm{rad/s^2}$ is recommended, as shown by the dotted lines in Fig. 10. However, because the slew rate is dependent on both parameters a and b and explicitly rewriting the maximum slew rate as a function of these inputs is not straightforward, it is recommended to use the contour plots in Fig. 6 as a visual guide to assist in the design of the DEC waveform.

Conclusion

In this study, a new chirp signal is proposed that is particularly useful for high-noise environments such as those encountered in orthogonal superposition rheometry. A high-frequency divergence or "blowout" was observed in traditional OWCh signals, and this is shown to be caused by a reduced SNR, especially at high frequencies, as verified with simulations of the linear Maxwell model. A new time-varying chirp waveform was designed by developing a new instantaneous frequency function $\omega_{dec}(t)$ and integrating the result to obtain the instantaneous phase function $\phi_{dec}(t)$. The instantaneous frequency function follows an S-shaped trajectory rather than a single exponential function, increasing the power spectral density at high frequencies and redistributing signal power across the frequency domain. The S-curve is constructed from two smoothly connected exponential functions to create a single continuous curve with the goal of retaining the exponential nature of the signal. Two new user inputs a and b were introduced to control the locus of the inflection point and fix the additional degrees of freedom in the double exponential curve (DEC). A range of optimal user inputs exists at low a and high b such that the slew rate $\alpha_{dec}(t)$ characterizing the rate of change in the instantaneous frequency at the end of the chirp pulse $\alpha_{dec}(T) < 1\,\mathrm{rad/s^2}$ and the SNR is maximized. Within this optimal region, asymptotic approximations for the parameters characterizing the DEC waveform are valid that simplify the waveform design. Orthogonal superposition rheometric measurements of a wormlike micellar solution were used to demonstrate the utility of the DEC waveform and for comparison against the OWCh. This double exponential chirp (DEC) successfully mitigates the high-frequency "blowout" effect particularly in high-noise environments such as those encountered in orthogonal superposition rheometry.

Specifically for chirp rheometry in noisy environments, the following steps can be taken to decide on the design and choice of the waveform. Firstly, the instrument sensitivity limit should be determined and compared to the values expected during the determination of the desired material property. If the expected value is within 1 to 2 orders of magnitude above this limit, in essence $G^*_{min} < G^* < 100\,G^*_{min}$, the DEC waveform is recommended. Secondly, the user should select appropriate values of a and b for the DEC waveform. From simulation results, $(a, b) = (0.5, 0.95)$ is recommended for initial experiments as this point lies within the optimal region and satisfies the criterion $a < a_{crit}$, discussed in (31). In general, the parameter b should be kept close to unity while a should be kept low to ensure the slew rate at the end of the chirp to be $\alpha_{dec}(T) < 1\,\mathrm{rad/s^2}$. The exact, optimal set of (a, b) values for each material might shift, and contour plots such as those provided in Fig. 6 can be helpful for identifying the best choices, but the optimal region as defined in this work is expected to remain generally valid.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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